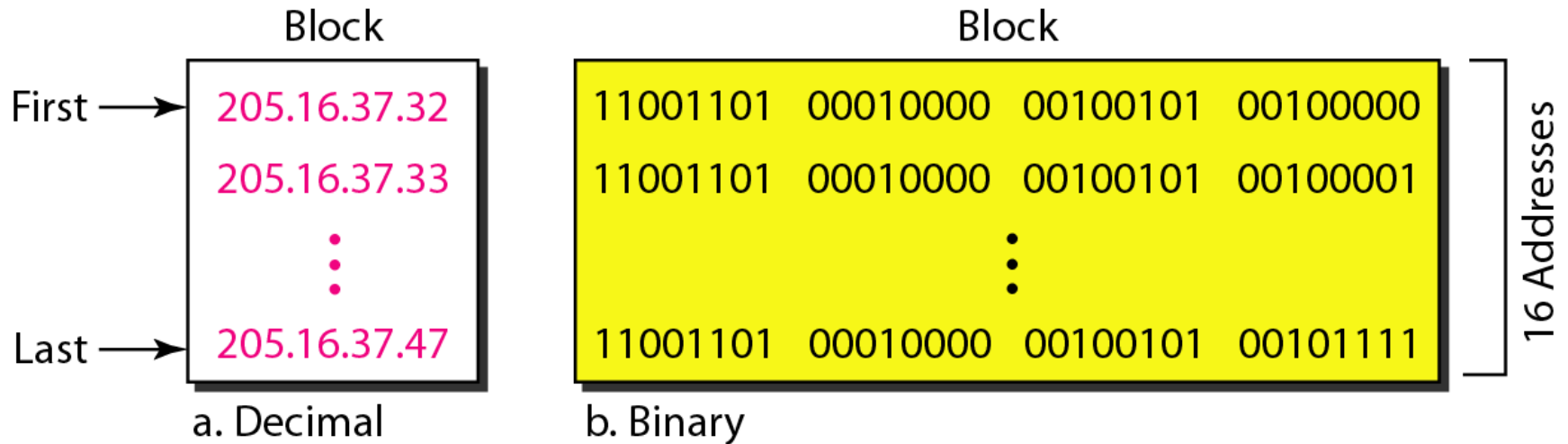


Network Layer: Logical Addressing

A block of 16 addresses granted to a small organization



This block of IP addresses is represented by:
205.16.37.32/28

Note

The first address in the block can be found by setting the rightmost $32 - n$ bits to 0s.

Example 1

A /28 block of addresses is granted to a small organization. We know that one of the addresses is 205.16.37.39. What is the first and last address in the block? What is its x.y.z.t/n representation?

Solution

The binary representation of the given address is

11001101 00010000 00100101 00100111

If we set 32–28 rightmost bits to 0, we get

11001101 00010000 00100101 00100000

or

205.16.37.32

The block representation is 205.16.37.32/28

Note

The last address in the block can be found by setting the rightmost $32 - n$ bits to 1s.

Example 1 (Continued)

last address for the block

Solution

The binary representation of the given address is

11001101 00010000 00100101 00100111

If we set 32 – 28 rightmost bits to 1, we get

11001101 00010000 00100101 00101111

or

205.16.37.47



Note

**The number of addresses in the block
can be found by using the formula
 2^{32-n} .**

Example 2

Another way to find the first address, the last address, and the number of addresses is to represent the mask as a 32-bit binary (or 8-digit hexadecimal) number. This is particularly useful when we are writing a program to find these pieces of information. In Previous example the /28 can be represented as

11111111 11111111 11111111 11110000

(twenty-eight 1s and four 0s).

Find

- a. The first address***
- b. The last address***

Example 2 (continued)

Solution

- a. The first address can be found by ANDing the given addresses with the mask. ANDing here is done bit by bit. The result of ANDing 2 bits is 1 if both bits are 1s; the result is 0 otherwise.

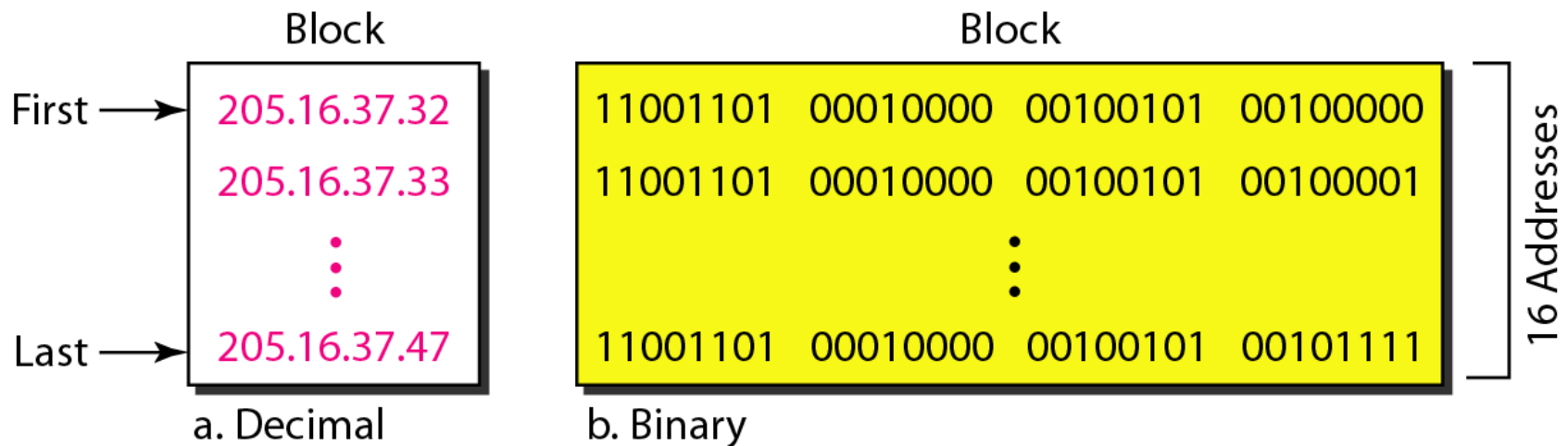
Address:	11001101	00010000	00100101	00100111
Mask:	11111111	11111111	11111111	11110000
First address:	11001101	00010000	00100101	00100000

Example 2 (continued)

- b. The last address can be found by ORing the given addresses with the complement of the mask. ORing here is done bit by bit. The result of ORing 2 bits is 0 if both bits are 0s; the result is 1 otherwise. The complement of a number is found by changing each 1 to 0 and each 0 to 1.*

Address:	11001101	00010000	00100101	00100111
Mask complement:	00000000	00000000	00000000	00001111
Last address:	11001101	00010000	00100101	00101111

A network configuration for the block 205.16.37.32/28



Example 2

An ISP is granted a block of addresses starting with 190.100.0.0/16 (65,536 addresses). The ISP needs to distribute these addresses to three groups of customers as follows:

- a. The first group has 64 customers; each needs 256 addresses.*
- b. The second group has 128 customers; each needs 128 addresses.*
- c. The third group has 128 customers; each needs 64 addresses.*

Assume the blocks of IPs are sequentially assigned. Design the subblocks and find out how many addresses are still available after these allocations.

Example 2 (continued)

Solution

Group 1

For this group, each customer needs 256 addresses. This means that 8 bits are needed to define each host. The prefix length is then $32 - 8 = 24$. The addresses are

<i>1st Customer:</i>	<i>190.100.0.0/24</i>	<i>190.100.0.255/24</i>
<i>2nd Customer:</i>	<i>190.100.1.0/24</i>	<i>190.100.1.255/24</i>
<i>...</i>		
<i>64th Customer:</i>	<i>190.100.63.0/24</i>	<i>190.100.63.255/24</i>
<i>Total = $64 \times 256 = 16,384$</i>		

Example 2 (continued)

Group 2

For this group, each customer needs 128 addresses. This means that 7 ($\log_2 128$) bits are needed to define each host. The prefix length is then $32 - 7 = 25$. The addresses are

<i>1st Customer:</i>	<i>190.100.64.0/25</i>	<i>190.100.64.127/25</i>
<i>2nd Customer:</i>	<i>190.100.64.128/25</i>	<i>190.100.64.255/25</i>
<i>...</i>		
<i>128th Customer:</i>	<i>190.100.127.128/25</i>	<i>190.100.127.255/25</i>
<i>Total = $128 \times 128 = 16,384$</i>		

Example 19.10 (continued)

Group 3

For this group, each customer needs 64 addresses. This means that 6 ($\log_2 64$) bits are needed to each host. The prefix length is then $32 - 6 = 26$. The addresses are

<i>1st Customer:</i>	<i>190.100.128.0/26</i>	<i>190.100.128.63/26</i>
<i>2nd Customer:</i>	<i>190.100.128.64/26</i>	<i>190.100.128.127/26</i>
<i>...</i>		
<i>128th Customer:</i>	<i>190.100.159.192/26</i>	<i>190.100.159.255/26</i>
<i>Total = $128 \times 64 = 8192$</i>		

Number of granted addresses to the ISP: 65,536

Number of allocated addresses by the ISP: 40,960

Number of available addresses: 24,576

An example of address allocation and distribution by an ISP

